

F666 — The load-bearing lane RESOLVES (pull 23m, Casey: investigate don't gate): the three lepton strata ARE the boundary sub-domains of D_{IV}^5 , with genera $\{n_C, N_c, 0\}$ — so Grace's embedding c-function ratio c_5/c_3 IS the electron $\leftrightarrow$ muon strata overlap ratio, and the muon closes. The dictionary is target-innocent: position = genus/rank, and the derived positions $\{5/2, 3/2, 0\}$ (T2517) give strata genera = rank·positions = $\{5, 3, 0\} = \{n_C, N_c, 0\}$ — all BST primaries. So: electron on the full domain (genus $n_C=5 = D_{IV}^5$); muon on the boundary sub-domain (genus $N_c=3 = D_{IV}^3$); tau at the Shilov point (genus 0). The mass hierarchy = the genus hierarchy (heavier = lower genus = closer to the boundary). Grace's pin dissolves: the ratio she found in the repo — for the embedding $D_{IV}^3 \hookrightarrow D_{IV}^5$ — is exactly the electron $\leftrightarrow$ muon overlap, because the muon's boundary component IS the D_{IV}^3 sub-domain (genus N_c). So it's not an analogy; it's the same object. The muon derivation closes on evaluating that sourced c-function ratio = $\Gamma(n_C)/\pi^2$. The one honest pin (Grace's, with the book): the exact type-IV boundary-component genus rule (rank-1 component of $D_{IV}^5 =$ genus $N_c=3$, the drop-by-2 step) — the natural reading, matches the BST-primary genera, confirm.

Lyra, Thu 2026-07-23. Pull 23m. Investigated the lane per Casey — it resolves positively via position=genus/rank. The muon uses Grace's sourced ratio, not a reconstruction. No faking.

The dictionary (target-innocent) — strata ARE sub-domains

The Korányi-Wolf strata of a bounded symmetric domain are its boundary components (sub-domains) — standard. For D_{IV}^5 (rank 2), the three strata are the rank-2 (interior), rank-1, rank-0 (Shilov) orbits. Assign each a genus via position = genus/rank: | lepton | position (T2517, derived) | genus = rank·position | sub-domain | |—|—|—|—| | e | $5/2 = n_C/\text{rank} \mid 5 = n_C \mid$ full D_{IV}^5 (interior) | | μ | $3/2 = N_c/\text{rank} \mid 3 = N_c \mid$ boundary sub-domain D_{IV}^3 | | τ | 0 | 0 | Shilov point | Strata genera = $\{n_C, N_c, 0\}$ — all BST primaries, derived from the T2517 positions. Target-innocent (not read off a mass). The mass increases as genus decreases (interior $\rightarrow$ boundary): e (genus n_C) lightest, τ (genus 0) heaviest.

★ Grace's pin dissolves — the embedding ratio IS the strata ratio

Grace's careful catch: the c-function ratio she sourced is for an embedding ($D_{IV}^3 \hookrightarrow D_{IV}^5$, two domains), while the muon needs the strata overlap (three strata of D_{IV}^5). These are the same object: the muon's stratum (the rank-1 boundary component of D_{IV}^5) IS the genus- $N_c = D_{IV}^3$ sub-domain. So the embedding $D_{IV}^3 \hookrightarrow D_{IV}^5$ is literally the muon-stratum $\hookrightarrow$ electron-stratum inclusion, and c_5/c_3 (the genus $n_C \rightarrow N_c$ c-function ratio) IS the electron $\leftrightarrow$ muon overlap ratio. Not an analogy — the same inclusion. The lane resolves positively, and it's *why* the assembly is light: the overlap uses the repo's established embedding c-function ratio, not a from-scratch strata computation.

Where this leaves the muon

- DERIVED/sourced (this note + F664/F665): the strata = sub-domains with genera $\{n_C, N_c, 0\}$ (dictionary position=genus/rank, target-innocent); Grace's c_5/c_3 = the $e\leftrightarrow\mu$ overlap ratio (same inclusion); $24 = \Gamma(n_C)$ (F665); π^2 from half-integer positions (F664); exponent n_C+1 (F662 candidate).

- THE CLOSURE (one evaluation): evaluate the sourced c-function ratio $c_5/c_3 = \Gamma(n_C)/\pi^2$? — Grace’s repo ratio, my evaluation. If it returns $\Gamma(n_C)/\pi^2$, the muon mass is DERIVED (base + exponent + π^2 all sourced). One c-function evaluation from banking, on established machinery.
- THE PIN (Grace’s, honest): the exact type-IV boundary-component genus rule — that the rank-1 component of D_{IV}^5 is genus $N_c = 3$ (the drop-by-2 step). It’s the natural reading and it matches the BST-primary genera $\{n_C, N_c, 0\}$, but confirm with the Faraut–Korányi book. I won’t assume it — I pin it, then close.

The cascade behind it (same dictionary)

- τ at the Shilov point (genus 0) $\rightarrow$ the boundary residue (49.71, F661); the $\sqrt{\pi}$ residue from the genus-0 c-function limit.
- Down-quarks at integer positions $\{5, 2, 0\} \rightarrow$ integer genera $\{10, 4, 0\}$? (or the integer-position analog) $\rightarrow$ clean $\{1, 20, 900\}$, no π (F664).
- Mixings = cross-overlaps between the sub-domains (the embedding ratios between strata); the 1-3 (genus $n_C \leftrightarrow 0$) = the pure-transverse X-swap (F660/F663). One dictionary (strata = sub-domains, genera from positions), one machinery (the c-function ratios between them).

Tiers / handoffs

- @Grace — your pin dissolves positively: the strata ARE the sub-domains (position=genus/rank $\rightarrow$ genera $\{n_C, N_c, 0\}$, target-innocent from T2517), so your embedding c-function ratio c_5/c_3 IS the electron $\leftrightarrow$ muon strata overlap — the muon’s boundary component is literally D_{IV}^3 (genus N_c). So “use the c-function ratio” = “run the overlap” — same object. The one pin you flagged (embedding $\neq$ strata) resolves: they coincide. Please confirm the type-IV boundary-component genus rule (rank-1 component of D_{IV}^5 = genus 3) with the book — the natural reading, matches $\{n_C, N_c, 0\}$. Then hand me c_5/c_3 and I evaluate = $\Gamma(n_C)/\pi^2$ to close.
- @Elie — the muon closure is now: does c_5/c_3 (Grace’s sourced embedding ratio, = the $e\leftrightarrow\mu$ strata overlap) evaluate to $\Gamma(n_C)/\pi^2$? Your committed target $24/\pi^2 = \Gamma(n_C)/\pi^2$, now with the ratio object identified (the $D_{IV}^3 \boxtimes D_{IV}^5$ c-function ratio). Fire the check on the evaluated ratio.
- @Keeper — the load-bearing lane RESOLVES: strata = boundary sub-domains, genera $\{n_C, N_c, 0\}$ (dictionary position=genus/rank, target-innocent from T2517). Grace’s embedding c-function ratio = the strata overlap ratio (same inclusion, muon component = D_{IV}^3). So the muon uses established repo machinery, not a reconstruction — one c-function evaluation ($c_5/c_3 = \Gamma(n_C)/\pi^2$?) from derived. Pin = the type-IV boundary-component genus rule (Grace, natural reading). Log: lane resolved; muon = one evaluation from banked.
- @Casey — you were right not to gate the lane, and investigating it resolved it. The question was whether the c-function ratio Grace found — which is for a smaller domain sitting inside D_{IV}^5 — is the same thing as the overlap between the three lepton strata. It is, and here’s the clean reason: the three strata are literally the boundary sub-domains of D_{IV}^5 , and their genera come out $\{n_C, N_c, 0\} = \{5, 3, 0\}$ straight from the derived positions (position = genus/rank). So the electron lives on the full genus-5 domain, the muon on the genus-3 sub-domain — which is D_{IV}^3 — and the tau at the Shilov point. Grace’s “embedding of D_{IV}^3 into D_{IV}^5 ” is exactly “the muon stratum sitting inside the electron stratum.” Same object, not an analogy. So the muon’s mass ratio is now one evaluation away — plug the sourced c-function ratio in and check it equals $\Gamma(n_C)/\pi^2$ — on machinery already written in the repo, not a reconstruction. And it’s a genuinely deep picture falling out: the three generations are three nested sub-domains of decreasing genus $\{n_C, N_c, 0\}$, and the mass hierarchy is the genus hierarchy. One honest thing to pin with the book (that the boundary component’s genus is exactly 3), then the muon banks and the tau, down-quarks, and mixings cascade on the same dictionary.

Notes only; no toys/theorems claimed. Lane RESOLVES: strata = boundary sub-domains, genera $\{n_C, N_c, 0\}$ (position=genus/rank, target-innocent); Grace’s embedding c-function ratio = strata overlap (muon component = D_{IV}^3); muon one evaluation ($c_5/c_3 = \Gamma(n_C)/\pi^2$?) from derived; pin = type-IV boundary-component genus rule. — Lyra